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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/152,569	01/10/2023	Jian-Hong LUO	AOIC027	4429

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APPLIED OPTOELECTRONICS, INC.
13115 JESS PIRTLE BLVD.
SUGAR LAND, TX 77478

EXAMINER	
TAVLYKAEV, ROBERT FUATOVICH	

ART UNIT	PAPER NUMBER
2896	

NOTIFICATION DATE	DELIVERY MODE
02/21/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/152,569	Applicant(s) LUO et al.	
	Examiner ROBERT TAVLYKAEV	Art Unit 2896	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-10 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-10 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/10/23 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

DETAILED ACTION

Claim Objections

3. Claims 2 is objected to because of the following informalities:

Claim 2 recites the limitations "of 400G optical module" and "of 800G optical module" which have typographical errors/omissions. For the purposes of this Action, the limitations are interpreted as "of a 400G optical module" and "of an 800G optical module". Appropriate corrections are required.

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

5. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

6. Claims 1, 3 – 6, 8, and 9 are rejected under 35 U.S.C. 103 as being unpatentable over Zhou et al (CN 114706175 A) in view of "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages" by Wang et al, Nature, vol. 562, pp. 103 – 112, 2018 (hereinafter Wang).

Regarding claim 1, Zhou discloses (Figs. 1 – 5; Abstract; para. 32 - 38) an optical module configuration, comprising:

a circuit board 3,7 comprising an accommodation through opening (with a substrate 7 disposed in the opening, as seen in Fig 3; claim 6); and

a transmitter optical subassembly 4-6,8-10, comprising a light emitting unit 5 (laser; para. 32) and a thin film lithium niobate (LiNbOx) modulator 10 (para. 32), the thin film LiNbOx modulator 10 being optically coupled with the light emitting unit 5 (via 6 and 8, as shown in Figs. 3, and 5; para. 33), and the thin film LiNbOx modulator 10 being disposed in the accommodation through opening (a shown in Figs. 1 and 2; Abstract; claim 6).

As seen in Figs. 1 and 2, the accommodation through opening allows placement of thin film LiNbOx modulators with different sizes up to a maximum size/footprint determined by a transverse size(s) of the accommodation through opening. While Zhou does not cite/exemplify a wide variety of thin film LiNbOx modulators supportive of different optical module specifications, such modulator variety is well known in the art. For example, Wang describes (Fig. 1; Abstract; pp. 101 – 102) a thin film LiNbOx modulator and provides examples (Table 1) of thin film LiNbOx modulators supportive of different optical module specifications (e.g., different 3dB electro-optic bandwidth of 45 GHz, 80 GHz, and 100 GHz in the first 3 rows of Table 1). It would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention that the accommodation through opening in Zhou can allows placement of various thin film LiNbOx modulators supportive of different optical module specifications, which are illustrated by Wang, in order to allow upgraded versions of thin film LiNbOx modulators with increasing 3dB bandwidths and modulation speeds. Such trend is noted by both Zhou (para. 2) and Wang (p. 103).

In light of the foregoing analysis, the Zhou – Wang combination teaches expressly or renders obvious all of the recited limitations.

Regarding claim 3, Zhou teaches that the circuit board 3,7 comprises a main body 3 comprising the accommodation through opening and a base 7 disposed in the accommodation

through opening, the light emitting unit 5 and the thin film LiNbOx modulator 10 is carried on the base 7 (as seen in Figs 1 and 2; claim 6). Since the accommodation through opening can be covered by plates 17,18 (as seen in Figs. 3 and 4), Zhou renders obvious that the base 7 has a smaller thickness than a total thickness and a total length of the accommodation through opening.

Regarding claim 4, Zhou teaches that the base 7 supports the laser 5 and its temperature may be controlled by a thermal electric cooler attached to a laser mount 4 which is disposed on the base 7. Zhou also teaches that the plate 18 (para. 32) is beneficial to heat dissipation of the laser 5. Hence, Zhou renders obvious that the base 7 can be made of a material with high thermal conductivity, such as a metal material.

Alternatively or additionally, the Examiner takers official notice that it is well known in the art of optical transceiver modules that their heat sinks are commonly made of metal in order to ensure high thermal conductivity and efficient heat dissipation.

Regarding claim 5, Zhou teaches that the accommodation through opening is non-coaxial with a central (longitudinal) axis of the circuit board 3, as seen in Figs. 1 and 2.

Regarding claim 6, the Zhou – Wang combination considers that the thin film LiNbOx modulator 10 can have a relatively short length, e.g., down to 5 mm (Extended Data Fig. 1c; 2nd para. of Section "Methods / Device fabrication" of Wang). Thus, the Zhou – Wang combination considers renders obvious that a length of the accommodation through opening can be on the same order, that is from several millimeters to a few centimeters in order to accommodate both the thin film LiNbOx modulator 10 and the laser source 5 (as shown in Figs. 9 and 10 of Zhou). Hence, the Zhou – Wang combination considers a range of transverse sizes of the

accommodation through opening that at least overlaps with the recited range and a prima facie case of obviousness exists (MPEP 2144.05).

It is also noted that (i) the transverse sizes of the opening depend on a particular application (a maximum speed of operation, etc); that (ii) the instant application does not provide any criticality for the exact values of the recited suzes; that (iii) it has been held that discovering the optimum or workable ranges of prior art involves only routine skill in the art (In re Aller, 105 USPQ 233); and that (iv) it has been held that "A recognition in the prior art that a property is affected by the variable is sufficient to find the variable result-effective." *In re Applied Materials'*, Inc., 692 F.3d 1289, 1297 (Fed. Cir. 2012). It is well settled that it would have been obvious for an artisan with ordinary skill to develop workable or even optimum ranges for result-effective parameters. *In re Boesch*, 617 F.2d 272, 276 (CCPA 1980); see also *In re Woodruff*, 919 F.2d 1575, 1577-78 (Fed. Cir. 1990). In this regard, the Zhou – Wang combination considers the transverse sizes of the opening a result-effective parameter than enables placement of the thin film LiNbOx modulator 10 and the laser source 5 of certain sizes.

Regarding claim 8, Zhou teaches (Figs. 1, 2, and 5 – 10) that the transmitter optical subassembly further comprises an output waveguide 11 (an optical fiber array 11; para. 32) disposed in the accommodation through opening (as seen in Figs. 1 and 2), and that the output waveguide 11 is optically coupled with a light propagation end of the thin film LiNbOx modulator 10 (as seen in Figs. 1, 2, and 5 – 10; para. 32 and 33).

Regarding claim 9, Zhou teaches (Figs. 1, 2, and 5 – 10) that the thin film LiNbOx modulator 10 comprises a light receiving end (receiving light form the laser 5) and a light propagation end (emitting light to an optical fiber array 11; para. 32) at the same side (left side of 10 in Figs. 2 and 5 – 10).

7. Claim 2 is rejected under 35 U.S.C. 103 as being unpatentable over Zhou in view of Wang, and further in view of "Optoelectronic Integrated Circuits for Growing Datacenters: Challenge, Strategy and Evolution" by Song et al, Smart Photonic and Optoelectronic Integrated Circuits XXI, Proc. of SPIE vol. 10922, paper 109220C, 2019 (hereinafter Song).

Regarding claim 2, Zhou teaches that the thin film LiNbOx modulator 10 is a first thin film LiNbOx modulator supportive of a 400G optical module/transceiver (Abstract; para. 4). While Zhou does not cite systems with higher transmission rates, Song discloses an optical system comprising optical modules/transceivers with optical modulators (Section 2.1 "Low driving-voltage modulators") and cites such modules "can be applied in the hyper scale datacenters, towards the 400G, 800G and even future 1.6T Ethernet applications" (last para. of Section 2.1; also Section V). It would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention that the optical module in Zhou could accommodate a second thin film LiNbOx modulator supportive of an 800G optical module, as another higher-bit-rate format cited by Song.

Wang describes (para. bridging columns on p. 103) a well-known fact that a high modulation speed can be achieved with a *shorter* electrode/device. Hence, the Zhou – Wang – Chen combination renders obvious that the second thin film LiNbOx modulator (enabling an 800G module) can have a smaller size (length) than the first thin film LiNbOx modulator.

8. Claims 7 and 10 are rejected under 35 U.S.C. 103 as being unpatentable over Zhou in view of Wang, and further in view of Reimer et al (US 12,032,211 B1).

Regarding claim 7, Zhou considers that the laser 5 is optically coupled to a light receiving end of the thin film LiNbOx modulator 10 via *free-space* optics 6 (Fig. 5) or direct butt-coupling (Figs. 9 and 10) and does not teach that optical coupling can alternatively be provided by an optical *waveguide*. However, Reimer discloses (Figs. 1 and 2; Abstract; 4:7 – 7:47) an input (low-confinement) waveguide 120 (5:11 – 19) that is optically coupled with a light receiving end of a thin film LiNbOx modulator 112,114 (4:50 – 52; "The low-confinement waveguide is optically coupled with the high-confinement waveguide" in the Abstract). It would have been obvious to a person of ordinary skill in the art before the effective filing date of the claimed invention that the light receiving end of the thin film LiNbOx modulator 10 can be optically coupled to the laser 5 via an input waveguide, in accordance to the teachings of Reimer. The motivation for using an input waveguide is that it can be configured as a *mode-size transformer* and thereby reduce coupling loss between two optical elements with different sizes of optical modes, such as a laser (or an optical fiber) and a thin film LiNbOx modulator (1:25 – 38).

The Zhou – Wang – Reimer combination considers that at least a portion (proximal to the thin film LiNbOx modulator 10) of the input waveguide is disposed in the accommodation through opening.

Regarding claim 10, the Zhou – Wang – Reimer combination considers that light into the thin film LiNbOx modulator 10 can be provided either from a laser array (para. 33 of Zhou) or an optical fiber array (as shown for an optical fiber 150 in Fig. 1 of Reimer). In the latter case, the transmitter optical subassembly further comprises a (input) fiber array disposed in the accommodation through opening, and the (input) fiber array is optically coupled with a light receiving end of the thin film LiNbOx modulator 10 (in similarity with the output fiber array 11 in Figs. 1, 2 and 5 of Zhou).

Conclusion

9. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

US 11,977,284 B2

10. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ROBERT TAVLYKAEV whose telephone number is (571)270-5634. The examiner can normally be reached 10:00 am - 6:00 pm, Monday - Friday.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, William Kraig can be reached on (571)272-8660. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

11. Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ROBERT TAVLYKAEV/

Primary Examiner, Art Unit 2896